PBL Project – Phase 2 Assignment

Belle – Training for a Career at TSMC

Your task is to investigate:

What does a person need to learn or study before working at TSMC or in the semiconductor industry?

1. Degree field
TSMC's own job postings consistently list backgrounds such as Materials Science, Engineering, Electrical Engineering, Chemical Engineering, Mechanical Engineering, Physics, Chemistry, or Optics as qualifying fields for equipment engineering roles. For manufacturing and IT-focused roles, TSMC lists Computer Science, Information Engineering, Industrial Engineering, Statistics, or Mathematics as relevant fields.

2. Programming and data skills
Basic programming languages such as C++ and SQL are commonly required. For more advanced data-focused roles, TSMC asks for experience with analytics tools like Python, R, SAS, Matlab, C/C++/C#, and Java, along with a background in applied statistics, operations research, data mining, and machine learning.

3. Technical process knowledge
For equipment-focused roles, TSMC looks for a basic technical understanding of IC processing equipment, along with basic knowledge of Statistical Process Control (SPC) and Design of Experiments (DOE) principles.

4. Work readiness
TSMC explicitly notes that new engineers must be willing to work in a fab manufacturing environment that operates 24x7.

5. Soft skills
Job listings summarizing TSMC-related roles note that candidates need strong problem-solving skills, teamwork, and attention to detail to succeed in the role.

For each important job, explain:

What should the person study?

Electrical Engineering, Materials Science, Physics, or Chemical Engineering — TSMC's own job postings usually require a **Master's degree or higher** for this role.

What skills do they need?

Deep understanding of semiconductor physics/manufacturing, ability to analyze process data, strong communication skills for working across cross-functional, geographically dispersed teams.

What training may happen after joining the company?

Learn the specific process flow (front-end, middle-end, back-end of line), shadow senior engineers, and work on yield/performance experiments.

Split into groups by area — trainees are grouped by process type: thin film, wet etch, dry etch, etc.

Practice on real machines — every classroom has the actual machines used on the real production line, not models

Photolithography classroom — has a real machine to practice on, plus a wafer transport cart

Learn from taken-apart machines — some equipment is disassembled so trainees can see how it works inside

Learn from the start — cameras inside the machines let trainees watch what happens to a wafer step by step

Your long-term goal could be to create a **"Semiconductor Employee Training Program"** for the exhibition.

TSMC Careers — Equipment Engineer (Etch):
https://careers.tsmc.com/en_US/careers/JobDetail/Equipment-Engineer-Etch/16963

TSMC Careers — Intelligent Manufacturing Engineer:
https://careers.tsmc.com/en_US/careers/JobDetail/Intelligent-Manufacturing-Engineer-IMC-MFG/5439

TSMC Careers — 2026 Campus Recruitment (Intelligent Manufacturing):
https://careers.tsmc.com/en_US/careers/JobDetail/2026-TSMC-Campus-Recruitment-Intelligent-Manufacturing-Engineer-IMC-MFG/19040**ZipRecruiter — TSMC Engineer Job**
https://www.ziprecruiter.com/Jobs/Tsmc-Engineer